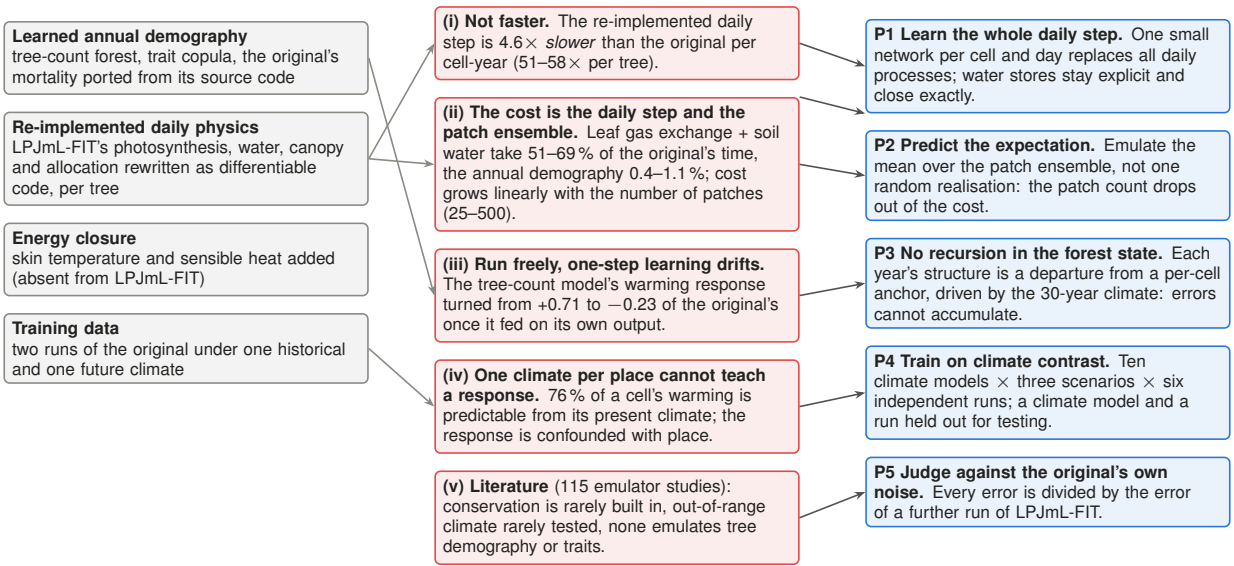
1. Starting point: a hybrid emulator
2. What the measurements showed
3. Design principles of the learned emulator
Learned annual demography
tree-count forest, trait copula, the original's mortality ported from its source code
Re-implemented daily physics
LPJmL-FIT's photosynthesis, water, canopy and allocation rewritten as differentiable code, per tree
Energy closure
skin temperature and sensible heat added (absent from LPJmL-FIT)
Training data
two runs of the original under one historical and one future climate
(i) Not faster. The re-implemented daily step is 4.6× slower than the original per cell-year (51–58× per tree).
(ii) The cost is the daily step and the patch ensemble. Leaf gas exchange + soil water take 51–69 % of the original's time, the annual demography 0.4–1.1 %; cost grows linearly with the number of patches (25–500).
(iii) Run freely, one-step learning drifts. The tree-count model's warming response turned from +0.71 to −0.23 of the original's once it fed on its own output.
(iv) One climate per place cannot teach a response. 76 % of a cell's warming is predictable from its present climate; the response is confounded with place.
(v) Literature (115 emulator studies): conservation is rarely built in, out-of-range climate rarely tested, none emulates tree demography or traits.
P1 Learn the whole daily step. One small network per cell and day replaces all daily processes; water stores stay explicit and close exactly.
P2 Predict the expectation. Emulate the mean over the patch ensemble, not one random realisation: the patch count drops out of the cost.
P3 No recursion in the forest state. Each year's structure is a departure from a per-cell anchor, driven by the 30-year climate: errors cannot accumulate.
P4 Train on climate contrast. Ten climate models × three scenarios × six independent runs; a climate model and a run held out for testing.
P5 Judge against the original's own noise. Every error is divided by the error of a further run of LPJmL-FIT.